deck_v4 presentation notes (business-school version)

Keyed to the frame numbers printed in the footline of `deck_v4.pdf`. Resynced on 5 October after David's own edits to the deck (a second "What a player sees" slide, Resignation moved after the screenshots, a new "Predictions" slide, shorter titles for the by-conflict slides). Main flow is 36 frames (title plus 35); 37 PDF pages because frame 19 (headline) has a two-step build, so PDF page numbers run one ahead of frame numbers from frame 20 on. Backup index is frames 37–38; FAQ index follows the backup slides. If a number here no longer matches, match on the quoted title.

Order of the main flow: 2 motivation; 3 why chess; 4–5 what a player sees; 6 resignation; 7 research questions; 8 predictions; 9 three conflicts; 10 data; 11 two samples; 12–14 descriptives; 15–16 estimation; 17 before onset; 18 after onset; 19 headline; 20 scale; 21 per-player versus per-game; 22 by conflict; 23 Armenia–Azerbaijan; 24 consistency across samples; 25 endings; 26–28 lost positions; 29 move quality; 30 openings; 31 specification curve; 32 limitations; 33 interpretation; 34 implications; 35 next steps; 36 conclusion. (The v3 numbers in David's comments of 3 October are mapped in `deck_v4_edit_log.md`.)

Slot: 60 minutes of talk followed by 20 minutes of questions (David, 5 October). The times given under each frame were written for a shorter slot and are minimums; they sum to about 33 minutes, so there is ample room to talk at length on every slide and nothing needs cutting or padding. A rough pacing for 60 minutes: motivation and background (frames 2–8) 11 min; data, descriptives and design (9–18) 17 min; headline, scale and per-player versus per-game (19–21) 7 min; results (22–31) 17 min; limitations, interpretation, implications, next steps, close (32–36) 8 min.

Rehearsal checkpoints at that pace: the event study done by minute 27; the headline (frame 19) by minute 30; persistence and quality done by minute 47.

Frame 2 — "Concession and the cost of continuing" (about 75 seconds)

Here is the economic angle. A dispute ends only when someone accepts an end to it: a settlement, a renegotiated loan, a disputed invoice paid or written off. Not conceding is costly, and we have good measurements of how costly. In wholesale vehicle negotiations, Brad Larsen estimates that failed trades destroy twelve to twenty-three percent of the available gains from trade. Roberts and Sufi find that most long-term credit agreements are renegotiated before maturity, so the willingness to adjust terms is the ordinary state of a credit relationship. And conflict reaches the institutions that settle disputes: after 2014, Russian courts enforced fewer claims by foreign parties.

Those studies measure the downstream consequences. Nothing is known about the decision process underneath: does a war change one person's willingness to concede to a particular counterpart from the other side, in an interaction that has nothing to do with the war? Online chess records that decision in its simplest form. Resignation is a unilateral concession, one click, directed at one identifiable opponent whose flag is on the screen. We observe it in thirty-two million games.

Reminders:

- Do not give the result here; it now comes on frame 19, after the design and the event study.
- If asked whether resignation is like a settlement: it is unilateral and ends the game as a loss, with nothing transferred; a settlement is two-sided with private information and transfers. The analogy is the decision to accept an end, not the game form.
- Citations: Larsen (2021) and Roberts and Sufi (2009) are cite-checked (both SUPPORTED; `docs/prose_drafts/cite_check_2026-10-03/`). The 12–23% is the union of bounds across two

seller types, relative to ex-post efficient gains from trade. Keep "long-term" with Roberts and Sufi: across all contracts the figure is 75%, over 90% only for maturities above a year.

Frame 3 — "Why chess?" (about 60 seconds)

Why this setting. People play online chess every day against strangers the platform pairs them with; the opponent's country is on the screen before the first move; and the game has nothing to do with the war. The same people meet opponents from the other side before the war and after it begins, far from the fighting. Almost no setting records such meetings at all; online chess records every one of them, move by move, with how the game ended. For the games analysed by an engine we also have an objective evaluation of every position, so we can separate what a player did from the situation they faced. Ratings put everyone on one scale, so like is compared with like. And chess has a long record as a measurement instrument in behavioural science.

If asked "chess players are not the public": agreed, and we make no claim about prevalence. The question is whether the same person changes how they treat opponents from a particular country once the countries are at war; that within-person change can be studied in any population that meets the other side before and after. Frame 10 has the sentence.

Frames 4–5 — "What a player sees" (two slides, about 60 seconds)

Frame 4 (David's slide): the live game search screen, where a player asks for a game and the platform finds an opponent. Frame 5: the opponent's name, rating and flag are on the screen before the first move and throughout. Pairing is by rating and availability; there is no way to filter opponents by country. The chat panel is there to show that the counterpart's country is part of the encounter; do not read it out, and do not draw conclusions from it.

Frame 6 — "Resignation: a unilateral concession" (45 seconds)

Either player can resign at any moment; the game ends immediately as a loss for the resigner. Etiquette treats resigning a clearly lost position as normal. Playing on in a lost position costs time and effort for little prospect of reward. For an economist: theory says a contest ends when one side decides that continuing costs more than it is worth (the war of attrition: two firms in a market that supports one, each losing money until one exits; Hendricks, Weiss and Wilson 1988). In practice people often do the opposite and keep investing in a losing course of action, which is escalation of commitment (Staw 1976: subjects who had chosen a failing investment put more money into it; meta-analysis in Sleesman et al. 2012). A lost chess position is a small version: each further move costs time and effort for almost no chance of saving the game. Two cautions if pressed: the analogy is only to the decision to stop paying, since the opponent bears no symmetric cost; and escalation of commitment is one of three readings, not a claimed mechanism. The platform records who conceded and when; for the engine-analyzed games, also how lost the position was, which is the expected value of continuing rather than conceding.

Frame 7 — "Research questions" (30 seconds)

State the question in the box, then the three questions. Name the three wars. Say once: we measure behaviour, not motives; treatment is based on the displayed flag, a proxy for national affiliation. If pressed, a flag is self-chosen and not citizenship; the FAQ slide has the stability audit (94% unchanged over seven years) and the direction of any misclassification (toward zero).

Frame 8 — "Predictions around participation and performance are not obvious" (David's slide)

David's own slide and wording; no script written. For reference when speaking to it, what the results later say about each pair of possibilities: playing less or more against the other side (contact fell, mostly through the supply of opponents; frames 13 and 32); refusing or seeking draws (no change in decisiveness or in any kind of draw; frame 25); playing worse or better (per-move error slightly lower, performance relative to ratings unchanged; frame 29); resigning more or less (less; frames 19–20).

Frames 9–11 — conflicts, data, two samples (about 2.5 minutes)

Frame 9: three wars, exact onset dates; the onset is the treatment and no player determines it. Frame 10: 32.4 million games, 33,687 accounts, selected on the flags they displayed before the war; hostile games are 1.5% of the corpus and the comparison set is the same player's other games. Chess players are not the public; the online population is very large; the estimand is a within-person change, which needs no claim about prevalence. Frame 11: two samples, blue and orange; the engine sample is the one that lets us condition on how lost a position was.

Frame 12 — "The raw-record sample by conflict" (about 45 seconds)

Who is in the data. 33,687 accounts and about 36 million player-game observations; an observation is one sampled player's side of one game. Ukraine–Russia is the largest conflict on both counts: 44% of accounts and 44% of observations. Armenia–Azerbaijan has 27% of accounts but 37% of observations, because its accounts play more; Israel–Palestine has 30% of accounts and 19% of observations. On average an account contributes about a thousand games, but that is a mean and activity is very unequal, which is why every result is reported per game and per player. Medians are not in the frozen outputs (asked of Eren).

Frame 13 — "How often players meet the other side" (about 60 seconds)

Hostile-country games are rare: about 2% of observations before onset, 3.2% in Ukraine–Russia and about half a percent in the other two conflicts. Everything else a player plays is the comparison set. The dark bars show the share after onset. It is lower, clearly so in Ukraine–Russia (3.19% to 2.41%). The main reason is supply: many Ukrainian players stopped playing in the first weeks of the war, and Russia blocked the platform on 23 April 2022. I come back to whether players also avoided these games under Limitations. Do not say the share "did not change", and do not quote the regression numbers from Eren's run (not adopted): raw fall about 18% pooled; after allowing for the other side's activity about 1% pooled per player-month, but 7% to 11% in Ukraine–Russia. Details: `docs/20261005_engagement_run_check.md`.

If asked which accounts the estimate comes from: every account is in the regression, but accounts with no hostile-country games only help pin down the month effects and rating controls. The estimate is driven by accounts with hostile-country games both before and after onset. In the raw records 18,852 accounts had such a game before onset and 14,151 after; the overlap is not in the frozen outputs. In the engine-analyzed sample 22,473 players have both, and restricting to them gives −0.46 pp against −0.45 pp for everyone (exploratory).

Units: these are shares of player-game observations, by conflict, from the frozen cell counts. They are not the average of each player's own share; that per-player figure is not in the frozen outputs.

If the selection question comes here, go to the backup slide "Ukraine–Russia, contact with the other side month by month" (linked from this slide and from Limitations) and say three things. First, there

is no drop at onset: the Ukrainian-side share steps down in May 2022, the first full month after the block, and is flat afterwards. Second, the Russian-side share dips in March, when many Ukrainians had stopped playing, and is back at its pre-war level from May to September; a game left before the first move disappears from both players' records, so avoidance would lower both series together. Third, we cannot observe such games, so avoidance is not excluded.

Spoken answers only, exploratory, engine-analyzed sample, not on any slide (`docs/20261005_contact_decline_chec`
in Ukraine–Russia the concession estimate is −0.60 pp (p = .036) between onset and the block and −0.28 pp (p = .15) afterwards, so it does not depend on the thinned pool; players lost hostile-country games as often as before (−0.12 pp, p = .54) and the rating gap moved by about one rating point; Israel–Palestine, where contact did not change, has the largest estimate. Do not claim the effect for both sides separately: by flag it is −0.49 pp (p = .026) for Ukrainian-flag players and −0.20 pp (p = .46) for Russian-flag players.

Frame 14 — "Hostile-country games and other games before the war" (about 45 seconds)

Before the war the two kinds of game were similar but not identical: the player resigned in 21.5% of hostile-country games and 20.3% of other games, and hostile-country games were about one move longer. That is fine. The design does not require the two to look alike; it requires the gap between them to have stayed the same had the war not begun, and the estimate is the change in that gap. This sets up the next slide. If asked about ratings or time controls by cell: not in the frozen outputs; ratings are controlled in every regression.

If asked whether the pre-war difference in resignation is significant: yes, but its sign depends on the comparison (not on the slide, at David's request). The table shows raw pooled means: 21.46% against 20.34%, a gap of +1.1 pp, far outside sampling error with 410,000 hostile-country observations. That raw gap is mostly composition: Ukraine–Russia supplies most hostile-country games and its players resign more in all their games. Within the same player, which is the comparison the design uses, hostile-country games had slightly FEWER resignations before the war: −0.25 pp (SE 0.08, p = .001; the Hostile coefficient in the headline regression, frozen). By conflict within player: Armenia–Azerbaijan −1.57 pp (p < .0001), Ukraine–Russia −0.17 pp (p = .049), Israel–Palestine +0.69 pp (p = .053). Within player, hostile-country games were also slightly longer (+0.17 moves, p < .0001). None of this is a problem for the design, which needs a stable gap, not a zero gap; the flat pre-onset bins are the evidence for that.

Frame 15 — "Estimation: the same player, two kinds of opponent, before and after" (about 90 seconds)

Point at the four cells. The estimate is how much the gap between the two rows changed after onset, within each player and within each calendar month of each conflict. Player fixed effects absorb everything stable about a player; conflict-by-month effects absorb activity waves and platform shocks; both ratings are controlled. The assumption, in one sentence: without the onset, the gap would have followed a similar path.

Frame 16 — "Estimation: the specification" (about 45 seconds)

The same comparison written out. The outcome is, for the main result, whether the focal player resigned. Hostile is one when the opponent displays the other belligerent's flag; Post is one after onset; beta on their interaction is the estimate, the change in the hostile-versus-other gap at onset within player and within month. Alpha is the player fixed effect; lambda is a conflict-by-calendar-month effect, which

also absorbs the post-onset main effect; both ratings are controlled; standard errors are clustered by player. A linear probability model, so effects read in percentage points. We report two weightings: each game equally and each player equally. For non-economists: this is the difference-in-differences from the previous picture; nothing more is going on. Do not linger; the next two slides show the assumption being checked.

Frames 17–18 — before and after onset (about 2.5 minutes)

Frame 17: each bar is the change in the gap in that three-month bin relative to the month before onset; zero is no change; the lines are confidence intervals. Before onset, nothing (jointly indistinguishable from zero). Frame 18: after onset, every bin negative; the dashed line is the full post-period estimate, which is the per-game number on the next slide; months six to eight sit near zero and the largest decline is in months nine to eleven; the display stops where the largest conflict leaves the window. Do not say the effect grows steadily.

If asked why the pre-onset bins need to be zero when the design already differences out the gap (FAQ slide, linked from frame 17, titled "Before onset: no pre-trend"; this is the pre-trends check for parallel trends): the design removes a difference in levels, not in trends. The level of the gap may differ from zero and does (within player, −0.25 pp before the war). What the bars show is the change in the gap relative to the month before onset, and that must be zero, because the one assumption is that the gap would have stayed put without the war. If the gap had already been drifting, the comparison of after with before would pick up the drift and call it an effect. The three pre-onset bins are −0.17, +0.20 and +0.11 pp (standard errors about 0.3 pp each), alternating in sign, jointly p = .32. Honest limits: not rejecting zero is not proof, since each bin is only as precise as a post-onset bin; and flat bins cannot rule out a shock exactly at onset specific to hostile pairings.

Frame 19 — headline (two builds, about 2 minutes)

Build 1. Here is the full statement of what the bins showed. Same player, their games against opponents showing the hostile country's flag versus their games against everyone else, after versus before the war began, with player and conflict-by-month fixed effects. Weighting every player equally, the typical player's probability of resigning a hostile-country game fell by one point four six percentage points, six point eight percent of the pre-war rate. The next slide puts a scale on that number.

Build 2. Counting every game equally instead, the decline is nought point four six percentage points, about two percent, about one fewer resignation in every hundred lost games; that is the dashed line on the previous slide. That average is dominated by the most active players. Both are negative; the gap between them is itself a finding, which comes after the scale slide. And it is not every war: Armenia–Azerbaijan shows no detectable change, which I come to.

Frame 20 — "How big is that? Measured against the games a player loses" (about 90 seconds)

Point at the bar: a thousand hostile-country games before the war, sorted by how they ended for the player.

A player almost always resigns only games they are losing, so the right yardstick is lost games, not all games. Out of a thousand hostile-country games a player lost about 460 by resignation, checkmate or on time: about 215 by resigning, 108 by checkmate and 135 on time. (A few more are lost in abandoned games; those are not in the raw-record file, so they sit in the grey segment.) So resignation ended about 47% of lost games. The estimate takes about 15 of the 215 away, leaving about 200. So 200 of the 460

lost games ended in resignation instead of 215: about 44% of lost games instead of 47%. Said a third way, 15 out of 460 is roughly three fewer resignations in every hundred lost games. The losses did not become rarer; they ended differently, which is the next section of the talk.

The hatched sliver on the bar is those 15 games. Weighting each game equally it is about 5 fewer, about 46% of lost games.

What the 6.8% is: the 1.46 pp fall divided by the pre-war LEVEL of resignation in hostile-country games (21.46%), not by the pre-war gap between hostile and other games. In words: before the war a player resigned about 21.5 of every 100 hostile-country games; judged against the same player's other games over the same months, that became about 20 after onset; 1.46 out of 21.46 is 6.8%, about one resignation in fifteen. The pre-war gap is a different and much smaller number (−0.25 pp within player), so the effect is several times the gap.

Units reminder: say "percentage points" for the estimate and "percent of the pre-war rate" for the relative figure; never mix them. The 6.8% does not depend on the yardstick (15 out of 215 either way). The 47% and 44% combine the per-player estimate with pooled baselines, the same convention the paper uses for the 6.8%. If asked whether losses changed: per game, resignations −0.46 pp, checkmated +0.22, lost on time +0.37, so these three together moved by about +0.1 pp.

Frame 21 — "Per-player vs per-game estimates" (about 90 seconds)

The per-game estimate weights each player by how much they play, so a threefold gap means the most active players changed least. The figure is the direct look: players grouped by how many hostile-country games they played before the war; the estimates shrink from the least to the most active quarter and span the two averages. The differences between quarters are imprecise, so the inference rests on the weighting comparison. The sentence that matters for this audience: occasional participants changed most. This is a comparison across players (less active players respond more), not an effect that fades within a player with repeated contact.

If asked "is it learning?": we tested it (exploratory, engine sample, not in the paper). We numbered each player's post-onset hostile games in order of play and let the effect vary by that order. It does not fade: the effect for a player's first hostile game after onset is no larger than for later ones (log-order interaction essentially zero, p = .88), and the profile is flat among heavy players too. Details: `docs/20261003_comment_tests_exploration.md`.

Frames 22–24 — by conflict, Armenia–Azerbaijan, consistency across samples (about 3 minutes)

Frame 22: Ukraine–Russia and Israel–Palestine decline, the latter imprecisely; Armenia– Azerbaijan no detectable change; dropping any one conflict leaves the pooled estimate negative. Frame 23: a possible explanation we cannot test in our data: a ceasefire about six weeks after onset. This conflict moves on a different margin, openings. If asked about recoding the post-ceasefire months: tested, exploratory; it leaves far too few hostile games to estimate anything and does not strengthen the pooled estimate. Frame 24: the separately built engine sample reproduces the per-game estimate almost exactly, and the per-conflict pattern and the weighting contrast; say "separately constructed", not "independent".

Frame 25 — "Where the missing concessions went" (about 90 seconds)

Each game counted once. Resignation endings fell by one and a half points; checkmates rose by nought point six five and losses on time by nought point eight, together one point four five. Games that would

have ended in concession ran on to checkmate or until a clock ran out. Draws and decisiveness did not change; games got slightly longer as a result. Say "where the endings moved, not why".

Frames 26–28 — lost positions (about 4 minutes)

Frame 26 (definition, the schematic): the engine evaluates every position; lost means three pawns or worse against the player, almost always decisive at these levels; the risk set is every game where that happened, and membership never depends on how the game ended. Two measures: half-moves after the position became lost, and continuing beyond six half-moves. Frame 27: plus nought point two eight eight half-moves per lost game, one point seven percent more, and more likely to continue beyond six half-moves. The cost falls on the persister. Frame 28 (robustness): the same games became lost (entry into the risk set unchanged), and the evaluation at which players resigned did not change, so they did not move their threshold.

Frame 29 — "Move quality: do players play worse?" (about 60 seconds)

No. The engine scores every move against its best move. The per-move error fell slightly; performance relative to the ratings did not change. Whatever kept players fighting, it was not accompanied by worse play. If asked about blunders: blunders per move fell slightly and mistakes per move did not change (exploratory, not in the paper); mistakes per game rose about one percent only because games are longer.

Frame 30 — openings (about 60 seconds)

Start with the first bullet for non-players: most players have a repertoire, a familiar set of first moves. Units: minus the log of how often the player had used that opening before the war; plus 0.1 is an opening used about ten percent less often. Pooled plus 0.087, 2.2% of the pre-war mean; Armenia–Azerbaijan plus 0.861, the biggest openings effect, in the conflict with no resignation change. Players departed from their habits but not toward more aggressive openings: the aggressive and safe opening shares did not change (pooled: aggressive −0.22 pp, p = .069; safe −0.04 pp; Armenia–Azerbaijan on its own: aggressive −0.41 pp with SE 0.38, safe −0.26 pp with SE 0.55, both far from significant). Do not say "less aggressive". If asked about gambits: an exploratory test using each opening code's pre-war draw rate shows no shift; a gambit classification would need Chess.com's opening names.

Frame 31 — "Sensitivity to analysis choices: specification curve" (about 60 seconds)

Fourteen already-reported specifications, every one negative. The two widest intervals drop the largest conflict; the two largest declines weight each player equally. The choice matrix and the dimensions table are one click away; do not go through them unless asked.

Frame 32 — "Limitations" (about 60 seconds)

Four points. A flag is not necessarily citizenship. Each account carries one recorded flag, taken before the war for focal players. We do NOT drop accounts that changed flag later, for three reasons: we have only two readings per account (collection and September 2024) and cannot date a change; dropping on a change that may itself be a response to the war (Russian-flagged accounts left that flag at twice the average rate) would select the sample on an outcome; and the per-account list is not yet in hand. The audit covers the whole database of 3.6 million accounts, not just our sample: 94.0% showed the same flag seven years on, 3.2% moved to another flag and 2.9% had no flag at the second reading. By country, any change: Russia 9.0%, Ukraine 6.1%, Israel 6.8%, Armenia 4.7%, Azerbaijan 4.8%. The direction of the resulting error is toward zero: an opponent recorded as Russian who was no longer showing

that flag makes an untreated game count as treated. The exception is an inflow into the Palestinian flag after 2023; that affects how some opponents are classified in the Israel–Palestine conflict only, and misclassification of that kind mixes untreated games into the treated set, which pulls the estimate toward zero. If asked whether we re-estimated without the switched accounts: not in the frozen outputs; the direction of the bias is what we can say. Players cannot filter by country, but who is online is not fixed. Aborts before move one are not stored, and abandonment inside stored games did not change. Contact with the other side fell after onset, most in Ukraine–Russia; the fall follows Russia's block of the platform on 23 April 2022 and not the onset, which points to supply; avoidance cannot be ruled out (backup slide and spoken answers: see the frame 13 note). Do not say "no selection effect". Early resignation (SI S10) is deliberately not raised; if a questioner raises it, the exploratory decomposition is in `docs/20261003_early_exit_exploration.md`.

Frames 33–36 — interpretation, implications, next steps, close (about 3.5 minutes)

Frame 33: three readings, consistent with the pattern, none adjudicated; what we can say is the real price; what is not certain includes the untested Armenia–Azerbaijan explanation. Frame 34: the transferable object is the decision to concede or accept an end; applications would need to record the route to an ending, the effort, and task quality separately; candidate settings and outcomes; more broadly, online strategic interactions between individuals. Say once that these are hypotheses and the chess estimates cannot be multiplied into money. Frame 35: Iran as a preregistered replication; the repeated-contact test; moderators; the structural companion work. Frame 36: the sentence in the box, then the three bullets, and stop.

Backup and FAQ

New backup slide (5 October): "Ukraine–Russia, contact with the other side month by month", listed under Dynamics in the second backup index and linked from frames 13 and 30. The two FAQ slides on avoiding opponents and quitting on seeing the flag were updated to match.

Backup index is two frames (data/design/specification; results in full, dynamics, heterogeneity), then the FAQ index. The estimating equation is now in the main flow (frame 16), so the backup no longer carries it. The v2 main-flow slides that moved to backup keep their content: Table 1, hostile games and the comparison set, identification in words with the platform quotes, the fixed-effects table, the threats table, both literature slides, the per-conflict forest in both samples, all endings and draws, all persistence measures, three quality measures, openings by colour, the specification curve with its matrix, and "is anyone driving it?". Robustness detail is one summary slide.

Next steps slide: a rematch test was added (supply of opponents held fixed; specification in the questions list for Eren, B6).